Inequalities — problem 7.75

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Prove that

$$x_i > 0 \text{ for all } i \Rightarrow x_1^{x_1} x_2^{x_2} \cdots x_n^{x_n} \geq (x_1 \cdots x_n)^{\frac{x_1+\cdots+x_n}{n}}.$$